

Stack Overflow Developer Survey Analysis

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OUTLINE



- Executive Summary
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- Methodology
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 - Visualization – Charts
 - Dashboard
- Discussion
 - Findings & Implications
- Conclusion

EXECUTIVE SUMMARY



- Tech Stack Continuity: High alignment between production tools and desired technologies indicates strong ecosystem stability
- Learning trends demonstrate a strategic pivot toward modern systems languages
- PostgreSQL is the #1 database across all OrgSize
- Generational Divide in Web Frameworks:
 - Senior developers (~12 yrs exp) maintain established enterprise stacks (ASP.NET, jQuery)
 - Junior developers (<8 yrs exp) lean toward modern JS/Python frameworks (Next.js, FastAPI)

INTRODUCTION



Summary of global developer profiles, current technology stack adoption, and future state of the global developer community from the Stack Overflow Developer Survey

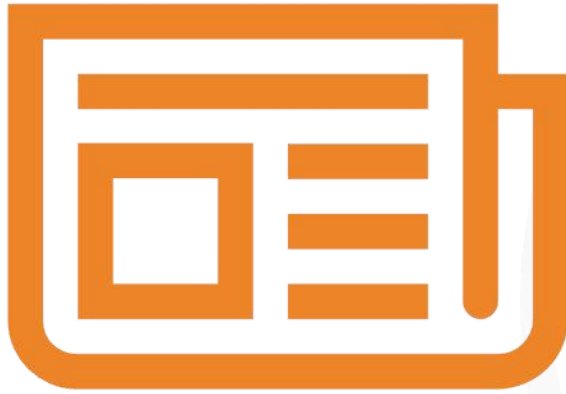
Target audience:

- Peer DA learners
- People with little technology knowledge

Takeaways:

- Industry understanding
- Future trends prediction

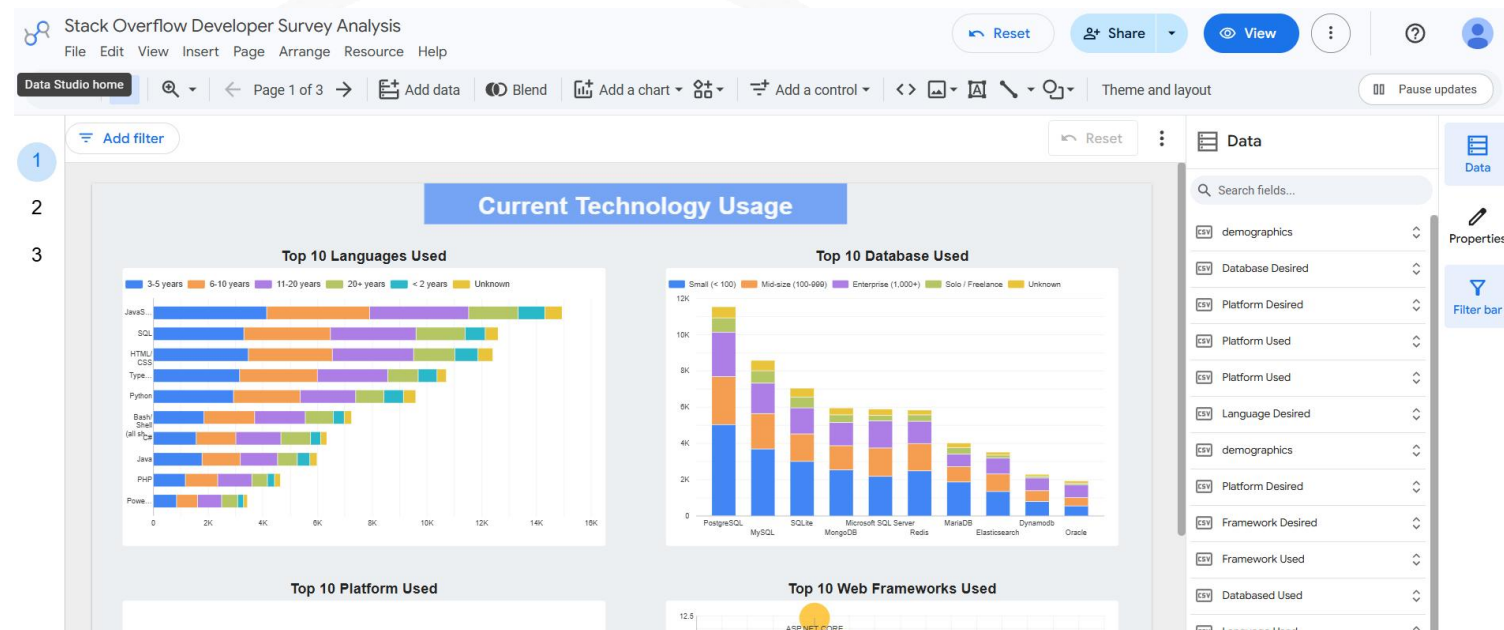
METHODOLOGY



Data sources, Data collection & Data wrangling steps

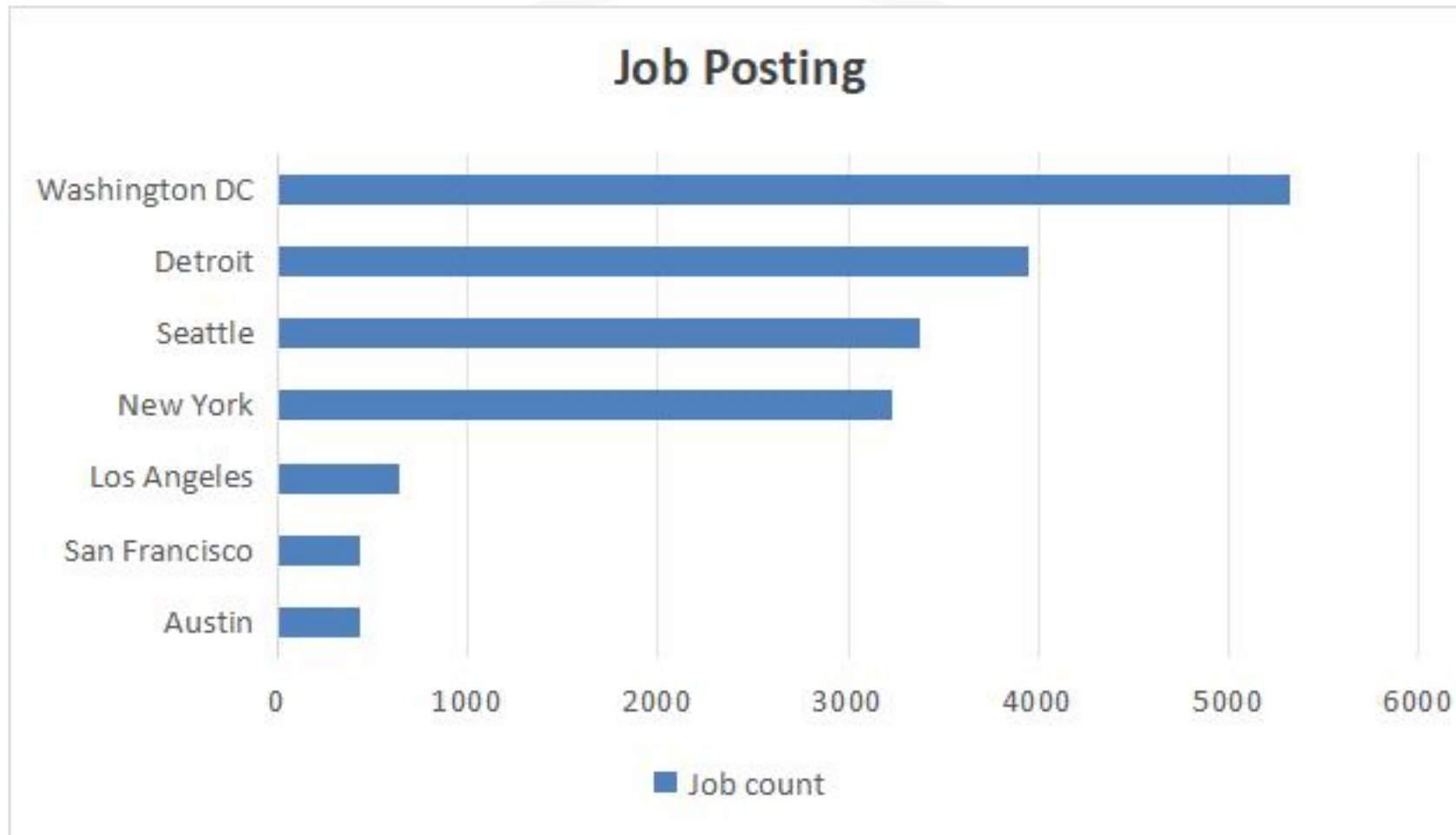
- The core dataset (survey_results_updated.csv) was extracted, cleaned, and processed using Python (Pandas)
- YearsCodePro, OrgSize columns are classified into tiers
- The data for visualization are stored in multiple CSV files
- Upload to Google Looker Studio for accurate aggregation and dashboard creation

Results

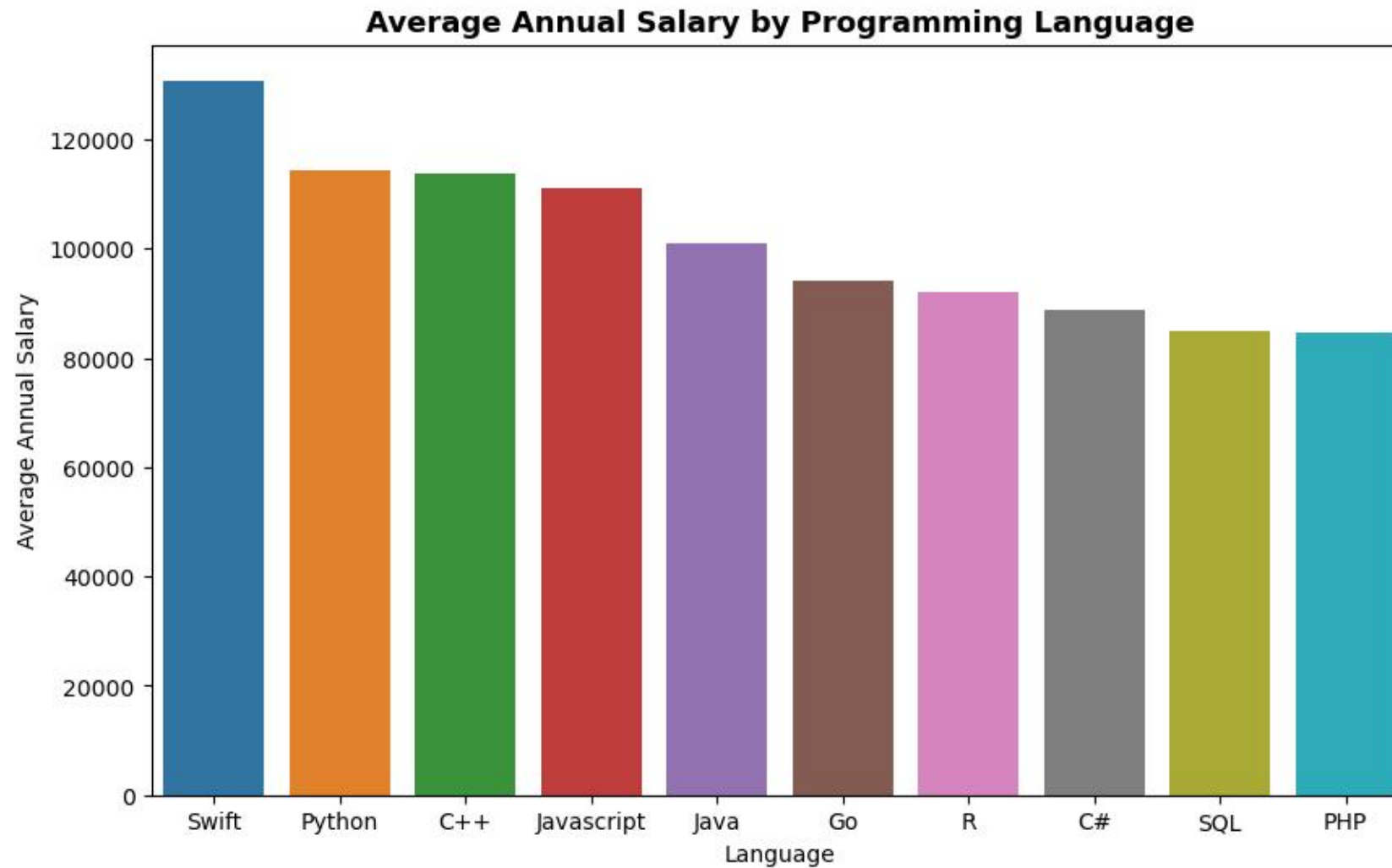


Shareable dashboard report

JOB POSTINGS



POPULAR LANGUAGES

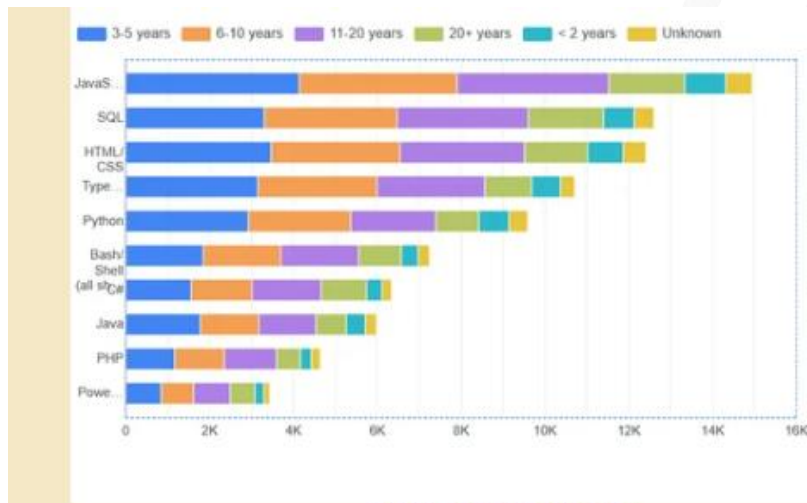


PROGRAMMING LANGUAGE TRENDS

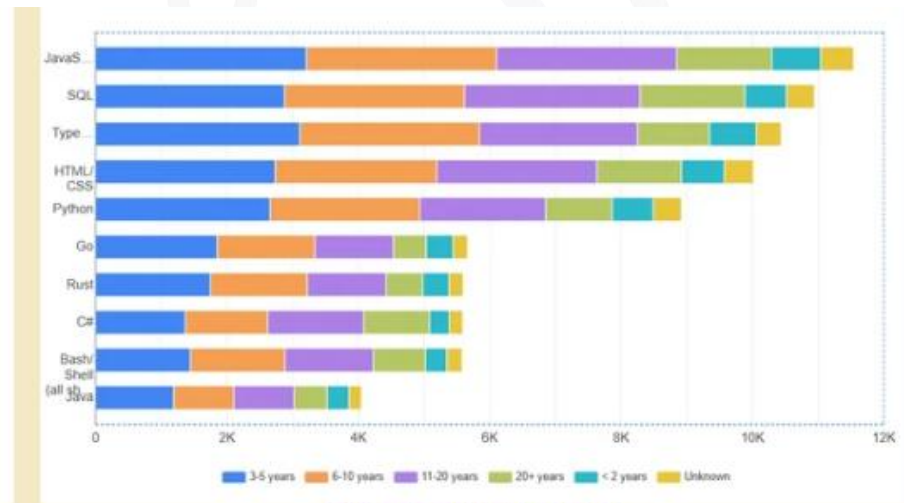
Both have similar in Language choice to learn and YearsCodePro distribution

Current Year

Next Year



TOP 10 LANGUAGE USED



TOP 10 LANGUAGE DESIRED



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- JavaScript, SQL, TypeScript, HTML/CSS, Python remain the core languages across both
- Rise of systems languages, such as Go, Rust, in the future trend
- Consistent Experience distribution: 3-10 years of experience make up the largest proportion across both

Implications

- Codebase Quality & Efficiency: The growing developer migration toward TypeScript, Go, and Rust reflects an industry-wide shift toward type safety, memory security, and runtime execution efficiency.



DATABASE TRENDS

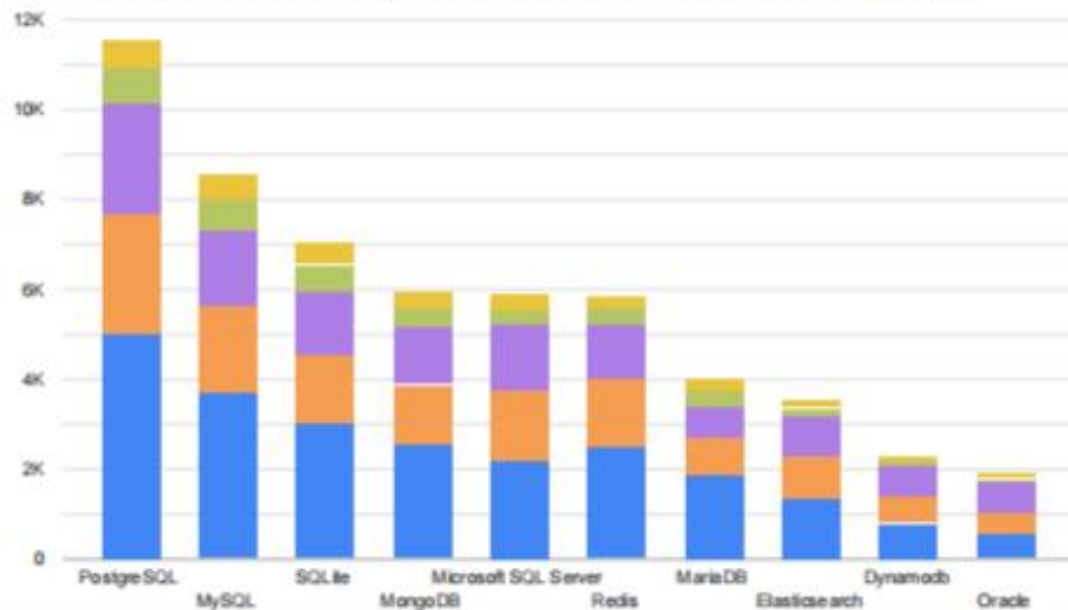
PostgreSQL remains #1 database, across OrgSize and the current and future

Current Year

Next Year

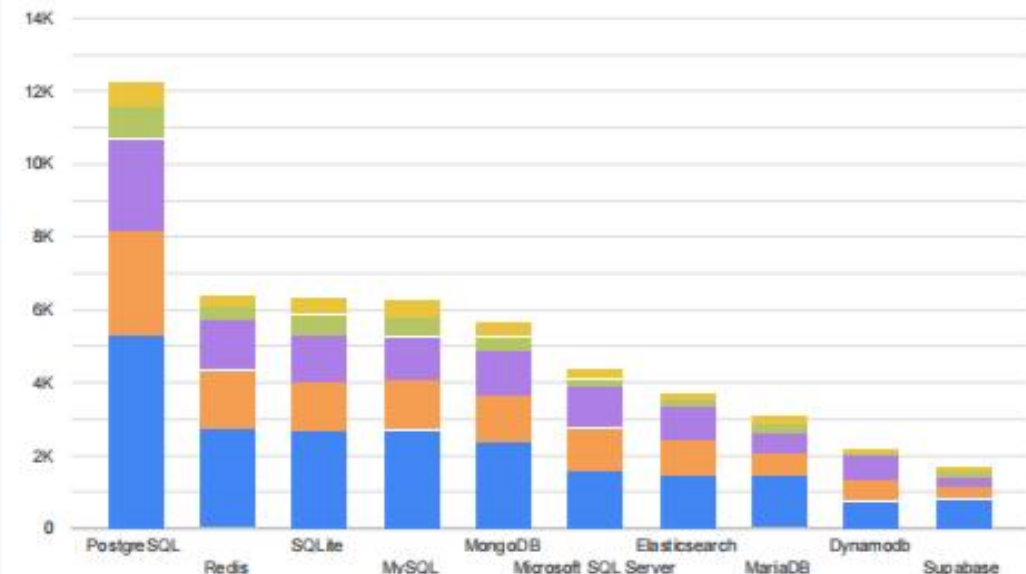
Top 10 Database Used

Small (< 100) Mid-size (100-999) Enterprise (1,000+) Solo / Freelance Unknown



Top 10 Database Desired

Small (< 100) Mid-size (100-999) Enterprise (1,000+) Solo / Freelance Unknown



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL expands its lead as the most desired database (~12,200)
- Microsoft SQL Server and Oracle skew toward large organizations (100+ employees), whereas the top databases have higher proportional usage in small companies and solo/freelance

Implications

- Since PostgreSQL dominates, companies should standardizing on it, unifies data pipelines
- Shift Toward High-Performance Caching & Developer-First Cloud Backends

DASHBOARD



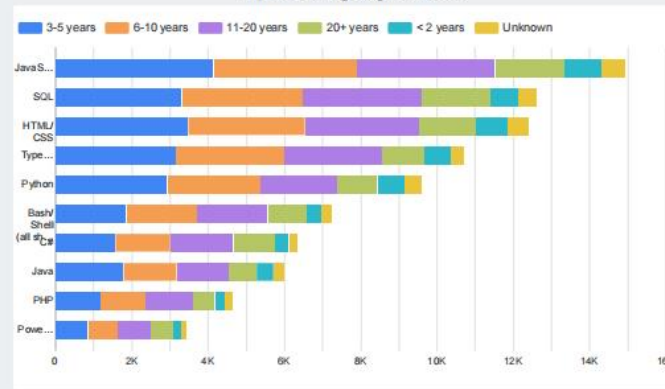
<Please present your dashboard in the following slides.>

DASHBOARD TAB 1: Current Technology Usage

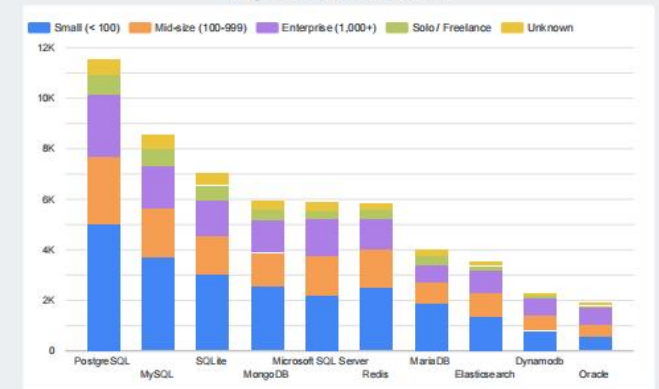
- JavaScript, SQL, HTML/CSS form core tech stack with devs 3-10 YearsCodePro
- React, Node.js lead overall adoption (~9000-9500)
- ASP.NET CORE, ASP.NET, and jQuery are maintained by more senior engineers

Current Technology Usage

Top 10 Languages Used



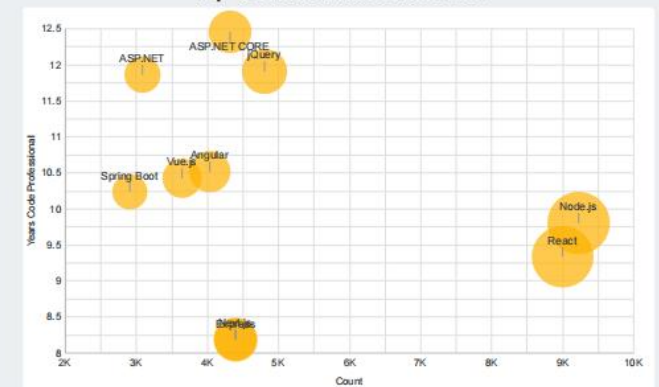
Top 10 Database Used



Top 10 Platform Used



Top 10 Web Frameworks Used

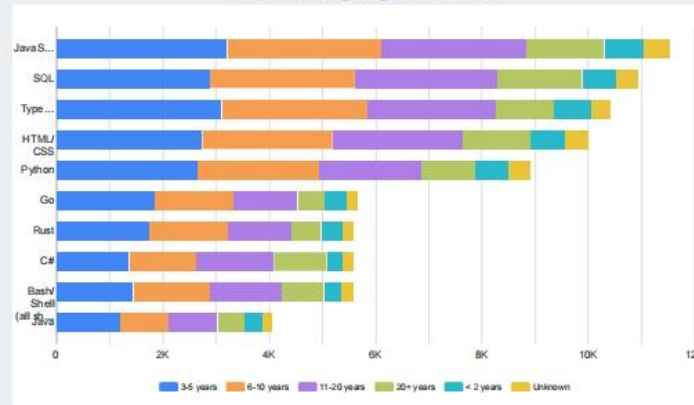


DASHBOARD TAB 2: Future Technology Trends

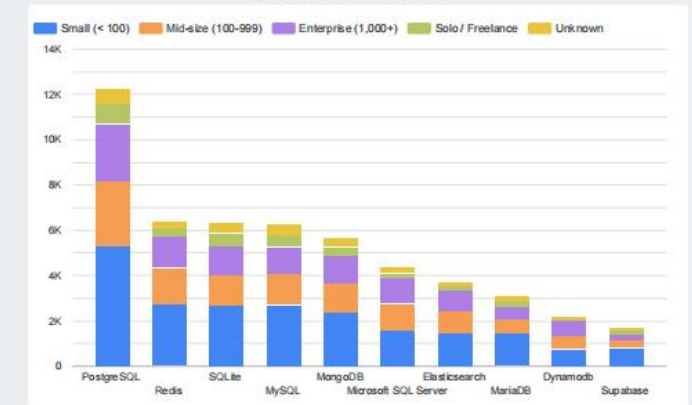
- Go and Rust surge into the top 10
- PostgreSQL expands its lead (~12,200)
- Developer interest in expanding toward specialized edge and deployment platforms like Vercel and Cloudflare

Future Technology Trend

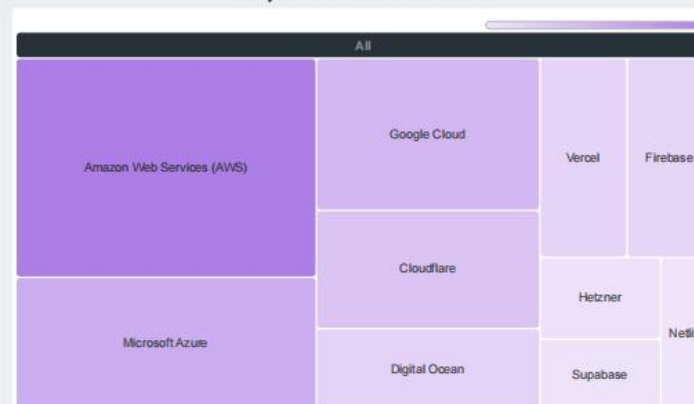
Top 10 Languages Desired



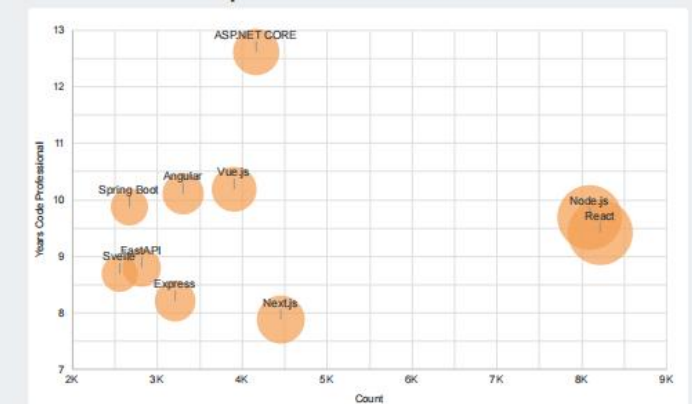
Top 10 Database Desired



Top 10 Platform Desired



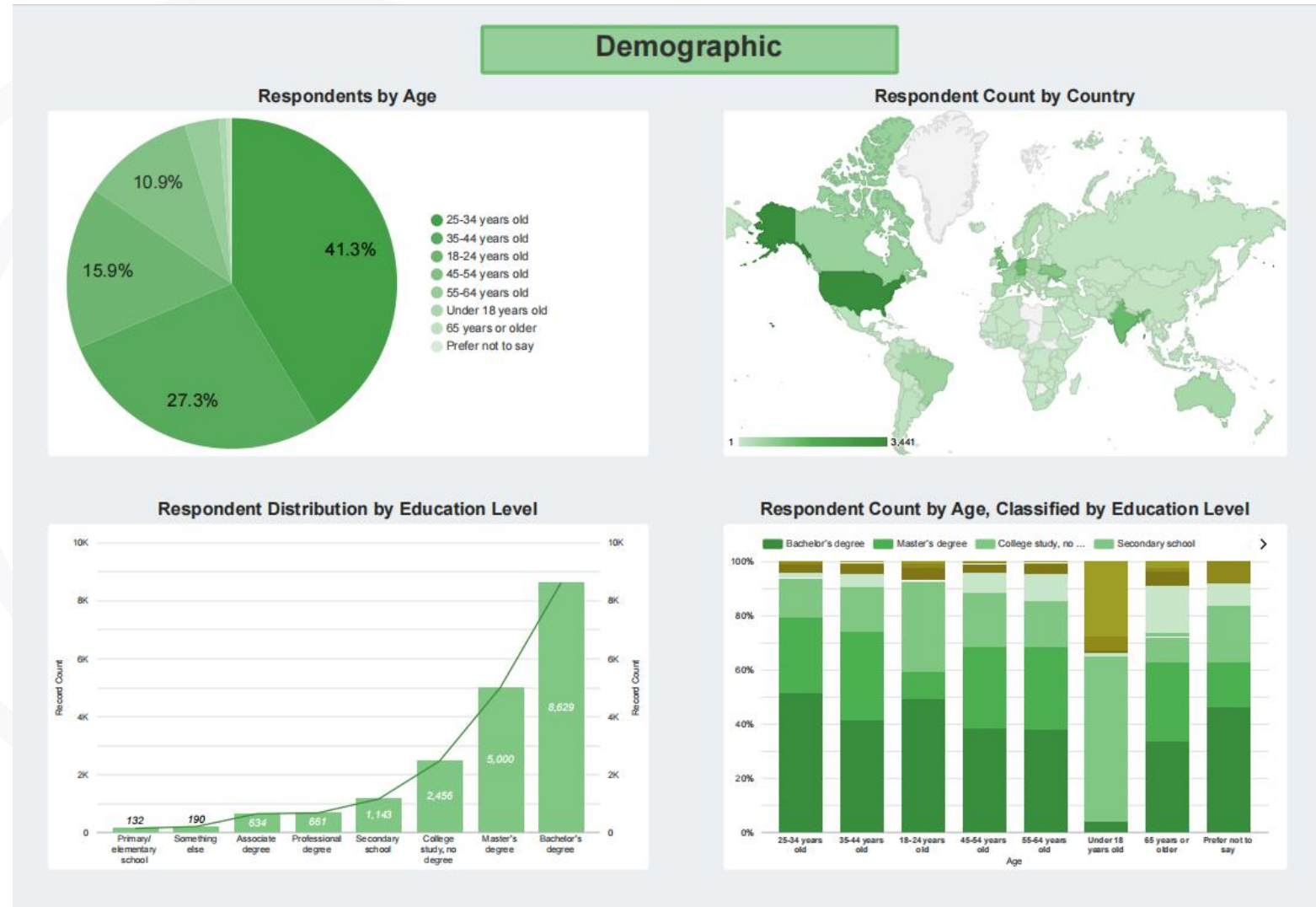
Top 10 Web Frameworks Used



DASHBOARD TAB 3: Demographics

Young and highly educated workforce demographics

- Early to mid-career:
 - 25-34 (41.3%)
 - 35-44 (27.3%)
 - 18-24 (15.9%)
- Majority are at college level
- United States represents the largest single country market



DISCUSSION



Overall:

- Tech stack continuity: core foundational tools—JavaScript, SQL, Python, PostgreSQL, and AWS continue to be the tools for devs in both the current and future
- Framework selection correlates directly with developer professional experience (YearsCodePro)
 - Seniors (averaging 11.8–12.5 YearsCodePro) - ASP.NET, jQuery
 - Juniors (averaging 8.0–8.5 YearsCodePro) - Next.js, FastAPI
- PostgreSQL serves as the universal database across every business size
- The "Big Three" (AWS, Azure, Google Cloud) handle core compute, while developer-first edge platforms (Vercel, Cloudflare, Supabase) capture rapid deployment workloads



OVERALL FINDINGS & IMPLICATIONS

Findings

- PostgreSQL is the #1 database across all organizational scales, while proprietary systems like Oracle and Microsoft SQL Server remain heavily restricted to large enterprises
- Web framework adoption is distinctly segmented by developer experience
- Modern compiled systems languages (Go, Rust) and static typing (TypeScript) represent the fastest-growing desired technologies

Implications

- Engineering organizations should standardize core data storage on PostgreSQL and align backend development with Go, Rust, and TypeScript
- Because the workforce is concentrated in the 18–34 age group, offering structured internal mobility and training in high-demand technologies (Rust, Go, Next.js) serves as a critical lever for engineering talent retention

CONCLUSION



- The global developer ecosystem remains supportive, with proven foundational standards (JavaScript, SQL, PostgreSQL) while actively modernizing toward high-performance systems languages (Go, Rust) and agile cloud-native frameworks.
- The engineering talent pool is young (68.6% under 35 years old) and academically grounded (72.1% hold university degrees), forming a dynamic and future-ready workforce that promises to drive sustained growth and innovation across the industry.
- Therefore, optimize engineering productivity and talent retention by standardizing core development on PostgreSQL and TypeScript/Go, upskilling teams in high-demand technologies like Rust and Next.js, take advantages of cloud infrastructure.



APPENDIX



- Include any relevant additional charts, or tables that you may have created during the analysis phase.